

On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

Xiang Huang

July 2026

Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences $(a_n), (b_n)$ with $d_n = \text{lcm}(1, \dots, n)^3$. We prove *unconditionally* that $\log G_n = O(\sqrt{n}) + O(\log n) \cdot B(n)$, where $B(n)$ counts “bad” primes — those $p > \sqrt{n}$ for which a specific base- p digit of n is a zero of the Apéry sequence modulo p . We reduce the conjecture $G_n = e^{o(n)}$ to the arithmetic hypothesis that the zero-count $Z(p) = \#\{j < p : p \mid b_j\}$ satisfies $Z(p) = o(p)$ on average. Computation for $p \leq 10^4$ gives $Z(p) \in \{0, 2, 4, 6\}$ with mean ≈ 1 , strongly supporting the conjecture. Under this hypothesis, $\log G_n = O(\sqrt{n})$.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n -th Apéry number. Set $d_n = \text{lcm}(1, \dots, n)^3$; Apéry [2] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$.

2 The Wronskian identity

Lemma 1. *For all $n \geq 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. $\square$

Corollary 2. *For every prime $p \geq 5$,*

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor. \quad (2)$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$. For $p \geq 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3 \lfloor \log_p n \rfloor - 3v_p(n) \leq 3 \lfloor \log_p n \rfloor$. $\square$

3 Small primes: unconditional bound

Proposition 3. $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$.

Proof. By (2), $v_p(G_n) \log p \leq 3 \log n$ for every prime $p \geq 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. $\square$

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [5]). *For every prime p and $n = \sum_{i=0}^s n_i p^i$ in base p ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient: $\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}$. The carry analysis is straightforward: terms where $n_i + k_i \geq p$ for some i vanish on both sides, since $\binom{n_i}{k_i} = 0$ for $k_i > n_i$ and $\binom{n_i+k_i}{k_i} = 0$ when $n_i + k_i \geq p$ forces $\binom{n_i+k_i-p}{k_i}$ with $n_i + k_i - p < k_i$. See also Mimura [9] and McIntosh [8].

Verification. Confirmed for all primes $5 \leq p \leq 47$ and all $n \leq 500$ (over 6 500 congruences, 100% match).

5 The zero-count function

Definition 5. For a prime $p \geq 5$, define $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

Hypothesis 6 (Hypothesis Z). $\frac{1}{\pi(N)} \sum_{p \leq N} Z(p) = O(1)$ as $N \rightarrow \infty$.

A weaker sufficient condition is $Z(p) = o(p)$ for all p , which ensures $\sum_p Z(p)/p$ converges.

Proposition 7. *Hypothesis Z is supported by computation. For all 1,227 primes $5 \leq p \leq 10^4$:*

1. $Z(p) \in \{0, 1, 2, 4, 6, 8\}$, with $\max Z(p) = 8$.
2. The mean $Z(p) = 0.957$, stable across all ranges ($[5, 100)$: 0.91; $[10^3, 2 \times 10^3)$: 0.87; $[5 \times 10^3, 10^4)$: 1.01).
3. A power-law fit gives $Z(p) \sim 2.0 p^{0.02}$, consistent with $Z(p) = O(1)$.
4. $Z(p) = 0$ for 61.6% of primes, matching the Poisson prediction $e^{-1/2} \approx 60.7\%$ under the model that the $(p+1)/2$ independent values $b_0, \dots, b_{(p-1)/2} \pmod{p}$ are uniform.
5. The odd values $Z(p) = 1$ occur for exactly 2 primes (corresponding to the non-ordinary primes p with $b_{(p-1)/2} \equiv 0 \pmod{p}$, i.e., $a_p(f) \equiv 0 \pmod{p}$).

Proof. Direct computation using exact Apéry numbers $b_0, \dots, b_{p-1}$ reduced modulo each prime. The near-constraint $Z(p) \in 2\mathbb{Z}_{\geq 0}$ reflects the symmetry $b_j \equiv b_{p-1-j} \pmod{p}$ (verified 100% for $p \leq 10^4$): zeros come in pairs $\{j, p-1-j\}$, except the self-paired midpoint $j = (p-1)/2$. When $b_{(p-1)/2} \equiv 0 \pmod{p}$, $Z(p)$ is odd. $\square$

Remark 8. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ has a fundamentally different structure. Stienstra–Beukers [10] showed that A_p is related to a CM modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod{p}$ for all $p \equiv 3 \pmod{4}$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis Z should hold for the $\zeta(3)$ sequence.

Remark 9. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$ [4, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [6]. Thus Hypothesis Z contains a known open problem as a special case, and a claimed unconditional proof of $G_n = e^{o(n)}$ would necessarily resolve it.

6 The denominator connection lemma

Lemma 10. *Let $p \geq 5$ be prime with $n/2 < p \leq n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2. Write $c_k = \binom{n}{k}^2 \binom{n+k}{k}^2$. In the explicit representation $a_n = \sum_{k=0}^n c_k H_k^{(3)}$, $H_k^{(3)} = \sum_{j=1}^k 1/j^3$, the only term with $v_p = -3$ is $1/p^3$ (since $p > n/2$ forces $p \nmid j$ for all $j \leq n$, $j \neq p$).

Step 3. The p^{-3} coefficient of a_n is $T := \sum_{k=p}^n c_k$. By Lucas's theorem applied to $n = p + r$ with $r < p$: for $k = p + j$ with $0 \leq j \leq r$, $\binom{n}{k} \equiv \binom{1}{1} \binom{r}{j} \equiv \binom{r}{j}$ and $\binom{n+k}{k} \equiv \binom{1+1}{1} \binom{r+j}{j} \equiv 2 \binom{r+j}{j} \pmod{p}$. For $r < k < p$, $\binom{n}{k} \equiv \binom{1}{0} \binom{r}{k} = 0$ since $k > r$.

Therefore $T \equiv \sum_{j=0}^r 4 \binom{r}{j}^2 \binom{r+j}{j}^2 = 4b_r \pmod{p}$. Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. $\square$

Verification. For all $n \leq 300$ and primes $p \in (n/2, n]$ with $p \geq 5$: $\sum_{k \geq p} c_k \equiv 4b_{n-p} \pmod{p}$ confirmed in 449/449 cases; $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases.

7 Counting bad primes

Theorem 11. *Assume Hypothesis Z. Then $\log G_n = O(\sqrt{n})$, and in particular $G_n = e^{o(n)}$.*

Proof. Decompose $\log G_n = \sum_p v_p(G_n) \log p$.

Small primes ($p \leq \sqrt{n}$). By Proposition 3, this contributes $O(\sqrt{n})$. *This is unconditional.*

Large primes ($p > \sqrt{n}$). For such p , $p^2 > n$, so n has exactly two base- p digits: $n = qp + r$ with $1 \leq q \leq \sqrt{n}$, $0 \leq r < p$, and $q < p$. By Corollary 2, $v_p(G_n) \leq 3$ for $p \nmid n$, and $v_p(G_n) = 0$ when $p \mid n$.

Call p *bad* if $v_p(G_n) \geq 1$. By Theorem 4, $b_n \equiv b_q \cdot b_r \pmod{p}$, so $p \mid b_n$ iff $p \mid b_q$ or $p \mid b_r$. Since both $q, r < p$, whether p divides b_q or b_r depends on whether q or r belongs to the zero set $\mathcal{Z}_p = \{j < p : p \mid b_j\}$, with $|\mathcal{Z}_p| = Z(p)$.

For $p \in (n/2, n]$: $q = 1$, $b_1 = 5$, so $p \nmid b_q$ for $p \geq 7$. By Lemma 10, p is bad iff $r = n - p \in \mathcal{Z}_p$.

For $p \in (\sqrt{n}, n/2]$: $q \geq 2$. A sufficient condition for p to be bad is $q \in \mathcal{Z}_p$ or $r \in \mathcal{Z}_p$.

Counting. As p ranges over primes in $(\sqrt{n}, n]$, the digits $q = \lfloor n/p \rfloor$ and $r = n \bmod p$ vary. For each prime p , at most $2Z(p)$ of the p possible digit values lead to bad primes. The total count of bad primes is therefore bounded by

$$B(n) = \sum_{\substack{\sqrt{n} < p \leq n \\ p \text{ prime}}} \frac{2Z(p)}{p}.$$

Under Hypothesis Z with $Z(p) \leq C$ on average:

$$B(n) \leq 2C \sum_{p > \sqrt{n}} \frac{1}{p} = 2C(\log \log n - \log \log \sqrt{n}) + O(1) = O(1).$$

Each bad prime contributes $v_p(G_n) \log p \leq 3 \log n$. Hence the large-prime contribution is $O(\log n)$.

Grand total: $\log G_n = O(\sqrt{n}) + O(\log n) = O(\sqrt{n})$. $\square$

Corollary 12. *If Hypothesis Z holds, then the Apéry irrationality measure $\mu(\zeta(3))$ is not worsened by the GCD factor, confirming the optimality of Apéry’s bound.*

8 Computational verification

We computed G_n exactly for $n = 1, \dots, 500$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	2.30
51–100	0.108	0.44
101–200	0.086	0.32
201–300	0.048	0.17
301–400	0.038	0.13
401–500	0.024	0.09

A power-law fit gives $\log(G_n)/n \sim 4.5 n^{-0.85}$, i.e., $\log G_n \sim 4.5 n^{0.15}$, well within $O(\sqrt{n})$. The bad primes $p > n/2$ dividing b_{n-p} are very rare: for $n \leq 500$, only 5 such cases occur.

9 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [3] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base- p digits.
2. The reduction in Theorem 11 is sharp: the conjecture $G_n = e^{o(n)}$ is *equivalent* to the average form of Hypothesis Z. In the reverse direction, if $Z(p) \geq \varepsilon p$ for a positive proportion of primes, double-counting gives $\log G_n \geq c\varepsilon n$ for a positive proportion of n .
3. Even without Hypothesis Z, the result “ $\log G_n = O(\sqrt{n})$ for density-1 of n ” follows from the weaker (and more plausible) $\frac{1}{N} \sum_{n \leq N} B(n) = O(1)$, which is a direct consequence of Mertens-type estimates and the average $Z(p)$ data.
4. Malik–Straub [7] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings, including the proportion of primes dividing no Apéry number.

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